

Total No. of Questions : 6]

SEAT No. :

P17

[Total No. of Pages : 3

APR-17/BE/Insem.-18

B.E. (Mechanical)

**REFRIGERATION AND AIR-CONDITIONING EQUIPMENT DESIGN
(2012 Pattern) (Elective - III(a)) (Semester - II)**

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer three questions out of 6.*
- 2) Solve Q1 or 2, Q3 or 4, Q5 or 6.*
- 3) All the three questions should be solved in one answer book and attach extra supplements if required.*
- 4) Draw diagrams wherever necessary.*
- 5) Use of scientific calculator is allowed.*
- 6) Assume suitable data wherever necessary.*

- Q1)** a) Explain with neat sketch the internal heat exchange cycle type transcritical refrigeration cycle. [5]
- b) Explain the performance characteristic curves of Reciprocating compressor. [5]

OR

- Q2)** What is dry ice? Explain with schematic diag. the method of manufacturing dry ice. [10]

- Q3)** a) Explain the construction working of pilot operated regulating valve. [5]
- b) Explain the construction working of direct acting solenoid valve. [5]

OR

- Q4)** a) Explain the construction working of direct acting solenoid valve. [5]
- b) Write a short note on "reverse cycle defrosting" [5]

P.T.O.

- Q5)** a) Explain the importance of Joule Thomson coefficient and inversion temperature. When operating a system for liquefaction of gases. [5]
b) Discuss specific types of insulations used for low temperature applications? [5]

OR

- Q6)** Determine the following for Linde-Hampson system with air as working fluid when the system is operated between 1 bar and 200 bar at 300k. [10]
a) Ideal work requirement
b) Liquid yield
c) Work per unit mass of compression
d) Work per unit mass of liquefaction
e) Figure of merit.

